

Main Project Using Ansible

Description

Instructions:

1. Install the necessary packages, which include MySQL 10.6, PHP version 8 or higher, and Nginx on your Linux server.
2. Create a user's home directory under the '/home' directory. You should set the root directory for the website to be '/home/username/website/public.'
3. Configure SFTP (Secure File Transfer Protocol) access for the user to ensure secure file management.
4. Configure phpmyadmin access for the user to ensure secure database management.
4. Install a free SSL certificate to enable secure HTTPS access to your website.
5. Once the website is hosted, proceed to the WordPress dashboard to make necessary configurations and customizations.
6. Write a personal blog post about yourself using the WordPress content management system.

Please share the following credentials after you finish the test.

Credentials:

- SFTP Credentials:

- Host:
- Username:
- Password:

- phpMyAdmin Credentials:

- URL:
- Username:
- Password:

- WordPress Credentials:

- WordPress Admin URL:
- Username:
- Password:

Please ensure that you document each step of the process and any configurations made. You may be asked to provide a brief report detailing the steps you took to complete this task.

1. Ansible Environment

The server configuration was automated using **Ansible**.

Two RHEL 10.2 EC2 instances were used:

System	Purpose
Control Node	Ansible Control Node
Target	Managed WordPress/Web Server

The Control Node contained:

```
ansible-core: 2.16.16
ansible-navigator: 26.9.0
Podman: 5.8.2
```

Ansible Navigator was installed inside a Python virtual environment.

The environment was activated using:

```
source ~/ansible-navigator-venv/bin/activate
```

The Ansible project was created with the following structure:

```
wordpress-project/
├─ ansible.cfg
├─ inventory
├─ site.yml
├─ group_vars/
│   └─ webservers.yml
└─ roles/
    └─ nginx/
        ├── tasks/
        │   └─ main.yml
        ├── handlers/
        │   └─ main.yml
        └─ templates/
```

```
|       └─ website.conf.j2
|
|─ php/
|   └─ tasks/
|       └─ main.yml
|
|─ mariadb/
|   └─ tasks/
|       └─ main.yml
|
|─ website_user/
|   └─ tasks/
|       └─ main.yml
|
|─ sftp/
|   └─ tasks/
|       └─ main.yml
|   └─ handlers/
|       └─ main.yml
|
|─ phpmyadmin/
|   └─ tasks/
|       └─ main.yml
|
|─ wordpress/
|   └─ tasks/
|       └─ main.yml
|   └─ templates/
|       └─ wp-config.php.j2
|
|─ ssl/
|   └─ tasks/
|       └─ main.yml
|   └─ handlers/
|       └─ main.yml
|   └─ templates/
|       └─ mywebsite.conf.j2
```

```
$ ssh -i "sanjid5.pem" ec2-user@ec2-98-94-99-237.compute-1.amazonaws.com
Register this system with Red Hat Lightspeed: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Tue Oct  6 09:58:28 2026 from 49.47.198.161
[ec2-user@ip-172-31-35-12 ~]$ source ~/ansible-navigator-venv/bin/activate
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 ~]$ ansible --version
ansible-navigator --version
podman --version
ansible [core 2.21.5]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/
share/ansible/plugins/modules']
  ansible python module location = /home/ec2-user/ansible-navigator-venv/lib64/pytho
n3.12/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansib
le/collections
  executable location = /home/ec2-user/ansible-navigator-venv/bin/ansible
  python version = 3.12.14 (main, Aug 13 2026, 00:00:00) [GCC 14.3.1 20251022 (Red H
at 14.3.1-4)] (/home/ec2-user/ansible-navigator-venv/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
ansible-navigator 26.9.0
podman version 5.8.2
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 ~]$ |
```

2. Ansible Configuration

ansible.cfg

The Ansible configuration was created as:

```
[defaults]
inventory = ./inventory
roles_path = ./roles
host_key_checking = False
interpreter_python = auto_silent
```

This configured Ansible to use the local inventory and the project's roles directory.

```
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 wordpress-project]$ cat ansible.cfg
[defaults]
inventory = ./inventory
roles_path = ./roles
host_key_checking = False
interpreter_python = auto_silent
```

inventory

The target server was added to the inventory:

```
[webservers]
target ansible_host=172.31.46.45 ansible_user=ec2-user
ansible_ssh_private_key_file=/home/ec2-user/.ssh/ansible_key
```

The private IP address was used because the Control Node communicated with the Target through the private network.

```
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 wordpress-project]$ cat inventory
[webservers]
target ansible_host=172.31.46.45 ansible_user=ec2-user ansible_ssh_private_key_file=/home/ec2-user/.
ssh/ansible_key
```

SSH Key

An Ansible SSH key was generated on the Control Node:

```
ssh-keygen -t ed25519 -f ~/.ssh/ansible_key
```

The key was used by Ansible to connect to the Target server.

3. Variables

The common variables were stored in:

```
group_vars/webservers.yml
```

The configuration was:

```
---
website_user: sanjid
website_name: mywebsite
website_root: "/home/{{ website_user }}/{{ website_name }}/public"

wordpress_db_name: wordpress
wordpress_db_user: wpuser
wordpress_db_password: "123"
wordpress_db_host: "localhost"
```

phpmyadmin_url_path: phpmyadmin

wordpress_admin_user: admin

wordpress_admin_password: "123"

wordpress_admin_email: "mhdsanjid07@gmail.com"

sftp_password: "[123]"

The variables allowed the roles to use the same website, database, and user configuration throughout the deployment.

```
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 wordpress-project]$ cat group_vars/webserver
.yml
---
website_user: sanjid
website_name: mywebsite
website_root: "/home/{{ website_user }}/{{ website_name }}/public"

wordpress_db_name: wordpress
wordpress_db_user: wpuser
wordpress_db_password: "123"
wordpress_db_host: "localhost"

phpmyadmin_url_path: phpmyadmin

wordpress_admin_user: admin
wordpress_admin_password: "123"
wordpress_admin_email: "mhdsanjid07@gmail.com"

sftp_password: "[123]"
```

4. Main Ansible Playbook

The complete deployment was controlled from:

site.yml

The final playbook was:

```
---
- name: Configure WordPress Server
  hosts: webserver
  become: true

  roles:
    - website_user
    - nginx
    - php
    - mariadb
    - sftp
```

- phpmyadmin
- wordpress
- ssl

The roles were executed in this order so that the website user and web server environment were prepared before WordPress and SSL configuration.

```
((ansible-navigator-venv) ) [ec2-user@ip-172-31-35-12 wordpress-project]$ cat site.yml
---
- name: Configure WordPress Server
  hosts: webservers
  become: true

  roles:
    - website_user
    - nginx
    - php
    - mariadb
    - sftp
    - phpmyadmin
    - wordpress
    - ssl
```

5. Website User and Directory Configuration

The `website_user` role was used to create the website user and directory structure.

roles/website_user/tasks/main.yml

```
---
- name: Create website user
  ansible.builtin.user:
    name: "{{ website_user }}"
    home: "/home/{{ website_user }}"
    shell: /bin/bash
    create_home: true
    state: present

- name: Create website directory
  ansible.builtin.file:
    path: "/home/{{ website_user }}/{{ website_name }}"
    state: directory
    owner: "{{ website_user }}"
    group: "{{ website_user }}"
    mode: "0755"
```

- name: Create website public directory
 ansible.builtin.file:
 - path: "{{ website_root }}"
 - state: directory
 - owner: "{{ website_user }}"
 - group: "{{ website_user }}"
 - mode: "0755"

- name: Set SELinux context for website directory
 community.general.sefcontext:
 - target: "{{ website_root }}(/.*)?"
 - setype: httpd_sys_content_t
 - state: present

- name: Apply SELinux context to website directory
 ansible.builtin.command:
 - cmd: "restorecon -Rv {{ website_root }}"
 changed_when: false

- name: Allow web services to access user home directories
 ansible.posix.seboolean:
 - name: httpd_enable_homedirs
 - state: true
 - persistent: true

- name: Allow traversal of website user's home directory
 ansible.builtin.file:
 - path: "/home/{{ website_user }}"
 - state: directory
 - mode: "0755"

The following directory was created:

/home/sanjid/mywebsite/public

This directory was used as the Nginx document root.

```
[ec2-user@ip-172-31-46-45 ~]$ ls -ld /home/sanjid
ls -ld /home/sanjid/mywebsite
ls -ld /home/sanjid/mywebsite/public
drwxr-xr-x. 4 sanjid sanjid 91 Oct 6 05:42 /home/sanjid
drwxr-xr-x. 3 sanjid sanjid 20 Oct 6 05:06 /home/sanjid/mywebsite
drwxr-xr-x. 6 sanjid sanjid 4096 Oct 6 06:26 /home/sanjid/mywebsite/public
[ec2-user@ip-172-31-46-45 ~]$ ls -la /home/sanjid/mywebsite/public
total 256
drwxr-xr-x. 6 sanjid sanjid 4096 Oct 6 06:26 .
drwxr-xr-x. 3 sanjid sanjid 20 Oct 6 05:06 ..
-rw-r--r--. 1 sanjid sanjid 180 Oct 6 05:12 index.html
-rw-r--r--. 1 sanjid sanjid 405 Oct 6 06:26 index.php
-rw-r--r--. 1 sanjid sanjid 20 Oct 6 05:17 info.php
-rw-r--r--. 1 sanjid sanjid 19903 Oct 6 06:26 license.txt
drwxr-xr-x. 12 sanjid sanjid 4096 Oct 6 06:18 phpmyadmin
-rw-r--r--. 1 sanjid sanjid 7407 Oct 6 06:26 readme.html
-rw-r--r--. 1 sanjid sanjid 7718 Oct 6 06:26 wp-activate.php
drwxr-xr-x. 9 sanjid sanjid 4096 Sep 22 12:55 wp-admin
-rw-r--r--. 1 sanjid sanjid 351 Oct 6 06:26 wp-blog-header.php
-rw-r--r--. 1 sanjid sanjid 2323 Oct 6 06:26 wp-comments-post.php
-rw-r--r--. 1 sanjid sanjid 3339 Oct 6 06:26 wp-config-sample.php
-rw-r--r--. 1 sanjid sanjid 372 Oct 6 06:26 wp-config.php
drwxr-xr-x. 4 sanjid sanjid 52 Aug 18 23:42 wp-content
-rw-r--r--. 1 sanjid sanjid 5617 Oct 6 06:26 wp-cron.php
drwxr-xr-x. 34 sanjid sanjid 12288 Sep 22 12:55 wp-includes
-rw-r--r--. 1 sanjid sanjid 2493 Oct 6 06:26 wp-links-opml.php
-rw-r--r--. 1 sanjid sanjid 3937 Oct 6 06:26 wp-load.php
-rw-r--r--. 1 sanjid sanjid 52536 Oct 6 06:26 wp-login.php
-rw-r--r--. 1 sanjid sanjid 8727 Oct 6 06:26 wp-mail.php
-rw-r--r--. 1 sanjid sanjid 33152 Oct 6 06:26 wp-settings.php
-rw-r--r--. 1 sanjid sanjid 35081 Oct 6 06:26 wp-signup.php
-rw-r--r--. 1 sanjid sanjid 5396 Oct 6 06:26 wp-trackback.php
-rw-r--r--. 1 sanjid sanjid 3205 Oct 6 06:26 xmlrpc.php
[ec2-user@ip-172-31-46-45 ~]$
```

6. Nginx Configuration

The nginx role installed and configured Nginx.

roles/nginx/tasks/main.yml

```
---
- name: Install Nginx
  ansible.builtin.dnf:
    name: nginx
    state: present

- name: Enable and start Nginx
  ansible.builtin.service:
    name: nginx
    state: started
    enabled: true

- name: Create Nginx virtual host configuration
```

```

ansible.builtin.template:
  src: website.conf.j2
  dest: /etc/nginx/conf.d/mywebsite.conf
  owner: root
  group: root
  mode: "0644"
notify: Restart Nginx

- name: Create test website
  ansible.builtin.copy:
    dest: "{{ website_root }}/index.html"
    content: |
      <!DOCTYPE html>
      <html>
      <head>
        <title>My Website</title>
      </head>
      <body>
        <h1>Welcome to My Website</h1>
        <p>This website was configured using Ansible.</p>
      </body>
      </html>
    owner: "{{ website_user }}"
    group: "{{ website_user }}"
    mode: "0644"

```

Nginx Template

The initial HTTP configuration was stored in:

```

roles/nginx/templates/website.conf.j2
server {
    listen 80;
    server_name _;

    root {{ website_root }};
    index index.php index.html;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }
}

```

```

location /phpmyadmin/ {
    try_files $uri $uri/ /phpmyadmin/index.php?$query_string;
}

location ~ ^/phpmyadmin/(.+\.php)$ {
    include /etc/nginx/fastcgi_params;
    fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
    fastcgi_param PATH_INFO "";
    fastcgi_pass unix:/run/php-fpm/www.sock;
}

location ~ \.php$ {
    include /etc/nginx/fastcgi_params;
    fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
    fastcgi_pass unix:/run/php-fpm/www.sock;
}
}

```

The Nginx service was enabled and started.

```

[ec2-user@ip-172-31-46-45 ~]$ nginx -v
nginx version: nginx/1.26.3
[ec2-user@ip-172-31-46-45 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/nginx.service.d
            └─php-fpm.conf
   Active: active (running) since Tue 2026-10-06 10:25:25 UTC; 4min 53s ago
 Invocation: d671d380f5f14008a8c9c4fe0265f732
    Process: 69340 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 69343 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 69345 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 69346 (nginx)
      Tasks: 3 (limit: 4156)
     Memory: 5.3M (peak: 5.6M)
        CPU: 62ms
    CGroup: /system.slice/nginx.service
            └─69346 "nginx: master process /usr/sbin/nginx"
              └─69347 "nginx: worker process"
                └─69348 "nginx: worker process"

Oct 06 10:25:24 ip-172-31-46-45.ec2.internal systemd[1]: Starting nginx.service - T
Oct 06 10:25:24 ip-172-31-46-45.ec2.internal nginx[69343]: nginx: the configuration
Oct 06 10:25:24 ip-172-31-46-45.ec2.internal nginx[69343]: nginx: configuration fil
[ec2-user@ip-172-31-46-45 ~]$

```

7. PHP and PHP-FPM Configuration

The php role installed PHP and the required extensions.

roles/php/tasks/main.yml

```
---
- name: Install PHP and required extensions
  ansible.builtin.dnf:
    name:
      - php
      - php-cli
      - php-common
      - php-fpm
      - php-gd
      - php-mbstring
      - php-mysqlnd
      - php-pdo
      - php-opcache
      - php-xml
      - php-json
      - php-curl
      - php-zip
    state: present

- name: Enable and start PHP-FPM
  ansible.builtin.service:
    name: php-fpm
    state: started
    enabled: true

- name: Create PHP test page
  ansible.builtin.copy:
    dest: "{{ website_root }}/info.php"
    content: |
      <?php
      phpinfo();
      ?>
    owner: "{{ website_user }}"
    group: "{{ website_user }}"
    mode: "0644"
```

PHP-FPM was configured to use:

/run/php-fpm/www.sock

The installed PHP version was:

PHP 8.3.33

```
[ec2-user@ip-172-31-46-45 ~]$ php -v
PHP 8.3.33 (cli) (built: Jul 28 2026 17:56:10) (NTS gcc x86_64)
Copyright (c) The PHP Group
Zend Engine v4.3.33, Copyright (c) Zend Technologies
with Zend OPcache v8.3.33, Copyright (c), by Zend Technologies
[ec2-user@ip-172-31-46-45 ~]$ sudo systemctl status php-fpm
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-10-06 05:17:11 UTC; 5h 13min ago
 Invocation: a113d499774c4a19a9c0ede6ddc7c2c2
    Main PID: 5021 (php-fpm)
   Status: "Processes active: 0, idle: 8, Requests: 344, slow: 0, Traffic: 0.00re"
    Tasks: 9 (limit: 4156)
  Memory: 112.8M (peak: 137.9M)
    CPU: 12.905s
   CGroup: /system.slice/php-fpm.service
           └─ 5021 "php-fpm: master process (/etc/php-fpm.conf)"
              5023 "php-fpm: pool www"
              5024 "php-fpm: pool www"
              5025 "php-fpm: pool www"
              5026 "php-fpm: pool www"
              5027 "php-fpm: pool www"
             29944 "php-fpm: pool www"
             38693 "php-fpm: pool www"
             43122 "php-fpm: pool www"

Oct 06 05:17:11 ip-172-31-46-45.ec2.internal systemd[1]: Starting php-fpm.service ->
Oct 06 05:17:11 ip-172-31-46-45.ec2.internal systemd[1]: Started php-fpm.service ->
[ec2-user@ip-172-31-46-45 ~]$
```

8. MariaDB Configuration

The assignment specified MySQL 10.6. For the actual implementation, **MariaDB 10.11.18** was used as the database server.

roles/mariadb/tasks/main.yml

- name: Install MariaDB server and Python MySQL library
 ansible.builtin.dnf:
 name:
 - mariadb
 - mariadb-server
 - python3-PyMySQL

```
state: present

- name: Enable and start MariaDB
  ansible.builtin.service:
    name: mariadb
    state: started
    enabled: true

- name: Create WordPress database
  ansible.mysql.mysql_db:
    name: "{{ wordpress_db_name }}"
    state: present
    login_unix_socket: /var/lib/mysql/mysql.sock

- name: Create WordPress database user
  ansible.mysql.mysql_user:
    name: "{{ wordpress_db_user }}"
    password: "{{ wordpress_db_password }}"
    priv: "{{ wordpress_db_name }}.*:ALL"
    host: "{{ wordpress_db_host }}"
    state: present
    login_unix_socket: /var/lib/mysql/mysql.sock
```

The database configuration was:

Database: wordpress
Username: wpuser
Host: localhost

The database was verified using:

```
sudo mariadb -e "SHOW DATABASES; SELECT User, Host FROM mysql.user;"
```

MariaDB version:

10.11.18

```
[ec2-user@ip-172-31-46-45 ~]$ mariadb --version
mariadb Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper
[ec2-user@ip-172-31-46-45 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-10-06 05:32:53 UTC; 4h 59min ago
 Invocation: de12b2720e9b441c9ce8a1ab1ed039c1
    Docs: man:mariabdd(8)
          https://mariadb.com/kb/en/library/systemd/
   Process: 11982 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 12007 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 12111 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR Galera=1; if [ $Galera = 1 ]; then /usr/bin/galera_recovery; fi (code=exited, status=0/SUCCESS)
   Process: 12125 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Process: 12156 ExecStartPost=/bin/rm -f /run/mariadb/wsrep-start-position (code=exited, status=0/SUCCESS)
  Main PID: 12113 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 4156)
    Memory: 134M (peak: 137.8M)
       CPU: 3.755s
    CGroup: /system.slice/mariadb.service
            └─12113 /usr/libexec/mariabdd --basedir=/usr

Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: you need to be root to run this command on Debian/Ubuntu derivatives
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: After creating the database directory, you need to
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: able to run the command as root.
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: See the MariaDB documentation for more details.
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: Please refer to the following URL:
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: The latest version of the MariaDB documentation is available at:
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: Consider upgrading to the latest version of the MariaDB documentation.
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal mariadb-prepare-db-dir[12063]: https://mariadb.com/kb/en/library/systemd/
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal (mariabdd)[12113]: mariadb.service: Reloaded successfully.
Oct 06 05:32:53 ip-172-31-46-45.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.11 database server.
```

```
[ec2-user@ip-172-31-46-45 ~]$ sudo mariadb -e "SHOW DATABASES; SELECT User, Host FROM mysql.user;"
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| wordpress |
+-----+
+-----+-----+
| User | Host |
+-----+-----+
| mariadb.sys | localhost |
| mysql | localhost |
| root | localhost |
| wpuser | localhost |
+-----+-----+
[ec2-user@ip-172-31-46-45 ~]$
```

9. SFTP Configuration

The sftp role configured secure file transfer access.

roles/sftp/tasks/main.yml

```
---
- name: Add control node SSH key for SFTP user
  ansible.posix.authorized_key:
    user: "{{ website_user }}"
    key: "{{ lookup('file', lookup('env', 'HOME') +
'/.ssh/ansible_key.pub') }}"
    state: present

- name: Set password for SFTP user
  ansible.builtin.user:
    name: "{{ website_user }}"
    password: "{{ sftp_password | password_hash('sha512') }}"
    update_password: always

- name: Ensure SFTP subsystem is configured
  ansible.builtin.lineinfile:
    path: /etc/ssh/sshd_config
    regexp: '^Subsystem\s+sftp'
    line: 'Subsystem sftp internal-sftp'
    state: present
  notify: Restart SSH

- name: Enable password authentication for SFTP
  ansible.builtin.lineinfile:
    path: /etc/ssh/sshd_config
    regexp: '^PasswordAuthentication'
    line: 'PasswordAuthentication yes'
    state: present
  notify: Restart SSH

- name: Override cloud-init password authentication setting
  ansible.builtin.lineinfile:
    path: /etc/ssh/sshd_config.d/50-cloud-init.conf
    regexp: '^PasswordAuthentication'
    line: 'PasswordAuthentication yes'
    state: present
  notify: Restart SSH

- name: Configure SFTP access for website user
  ansible.builtin.blockinfile:
```

```
path: /etc/ssh/sshd_config
marker: "# {mark} ANSIBLE SFTP CONFIGURATION"
block: |
    Match User {{ website_user }}
        ForceCommand internal-sftp
        X11Forwarding no
        AllowTcpForwarding no
        PermitTunnel no
notify: Restart SSH
```

SFTP Handler

```
---
- name: Restart SSH
  ansible.builtin.service:
    name: sshd
    state: restarted
```

Initially, password authentication was disabled by the Cloud-Init configuration.

This was identified using:

```
sudo sshd -T | grep '^passwordauthentication'
```

The result initially showed:

```
passwordauthentication no
```

The Ansible role was then updated to override the Cloud-Init setting.

After applying the playbook:

```
sudo sshd -T | grep '^passwordauthentication'
```

returned:

```
passwordauthentication yes
```

SFTP Test

SFTP was tested using:

```
sftp -o PreferredAuthentications=password -o PubkeyAuthentication=no
sanjid@52.87.201.167
```

The connection was successful:

Connected to 52.87.201.167.
sftp>

The remote directory was:

/home/sanjid

The website directory was successfully accessed:

/home/sanjid/mywebsite/public/

The website files, WordPress files and phpmyadmin directory were visible.

```
[ec2-user@ip-172-31-46-45 ~]$ sftp sanjid@52.87.201.167
sanjid@52.87.201.167's password:
Connected to 52.87.201.167.
sftp> pwd
Remote working directory: /home/sanjid
sftp> cd /home/sanjid/mywebsite/public/
sftp> ls
index.html          index.php           info.php
license.txt         phpmyadmin          readme.html
wp-activate.php     wp-admin            wp-blog-header.php
wp-comments-post.php wp-config-sample.php wp-config.php
wp-content          wp-cron.php         wp-includes
wp-links-opml.php   wp-load.php         wp-login.php
wp-mail.php         wp-settings.php     wp-signup.php
wp-trackback.php    xmlrpc.php
sftp> |
```

10. phpMyAdmin Configuration

The phpmyadmin role installed phpMyAdmin.

roles/phpmyadmin/tasks/main.yml

```
---
- name: Install phpMyAdmin dependencies
  ansible.builtin.dnf:
    name:
      - php-mbstring
      - php-mysqldb
      - php-json
      - php-xml
      - php-gd
```

- php-fpm
- unzip
- tar

state: present

- name: Create phpMyAdmin directory
 - ansible.builtin.file:
 - path: "{{ website_root }}/phpmyadmin"
 - state: directory
 - owner: "{{ website_user }}"
 - group: "{{ website_user }}"
 - mode: "0755"

- name: Download phpMyAdmin
 - ansible.builtin.get_url:
 - url: "<https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz>"
 - dest: "/tmp/phpmyadmin.tar.gz"
 - mode: "0644"

- name: Extract phpMyAdmin
 - ansible.builtin.unarchive:
 - src: "/tmp/phpmyadmin.tar.gz"
 - dest: "/tmp"
 - remote_src: true

- name: Find extracted phpMyAdmin directory
 - ansible.builtin.find:
 - paths: /tmp
 - patterns: "phpMyAdmin-*-all-languages"
 - file_type: directory
 - register: phpmyadmin_dirs

- name: Copy phpMyAdmin files to website
 - ansible.builtin.copy:
 - src: "{{ phpmyadmin_dirs.files[0].path }}"
 - dest: "{{ website_root }}/phpmyadmin/"
 - remote_src: true
 - owner: "{{ website_user }}"
 - group: "{{ website_user }}"
 - mode: "0755"

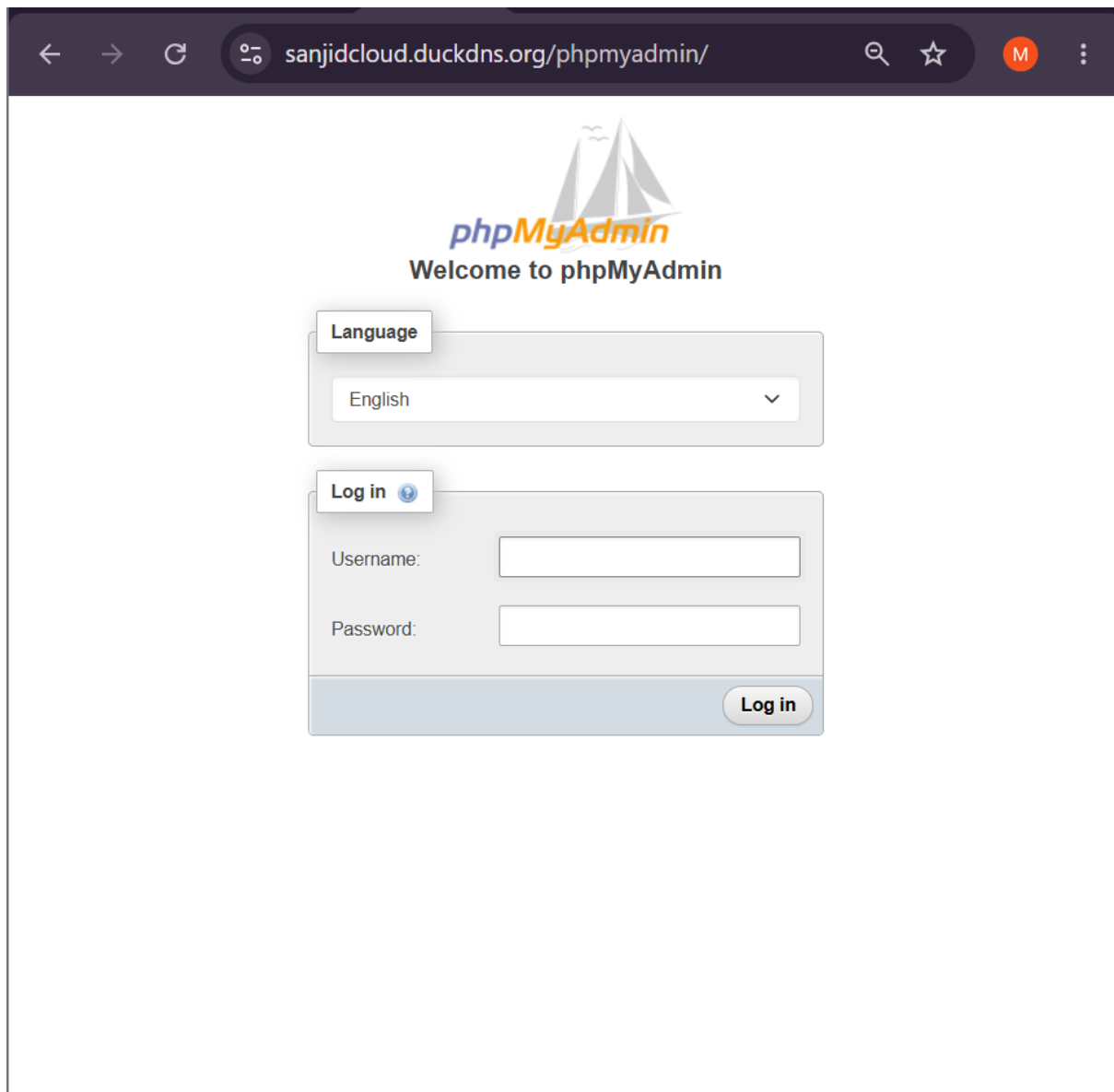
phpMyAdmin was installed at:

/home/sanjid/mywebsite/public/phpmyadmin/

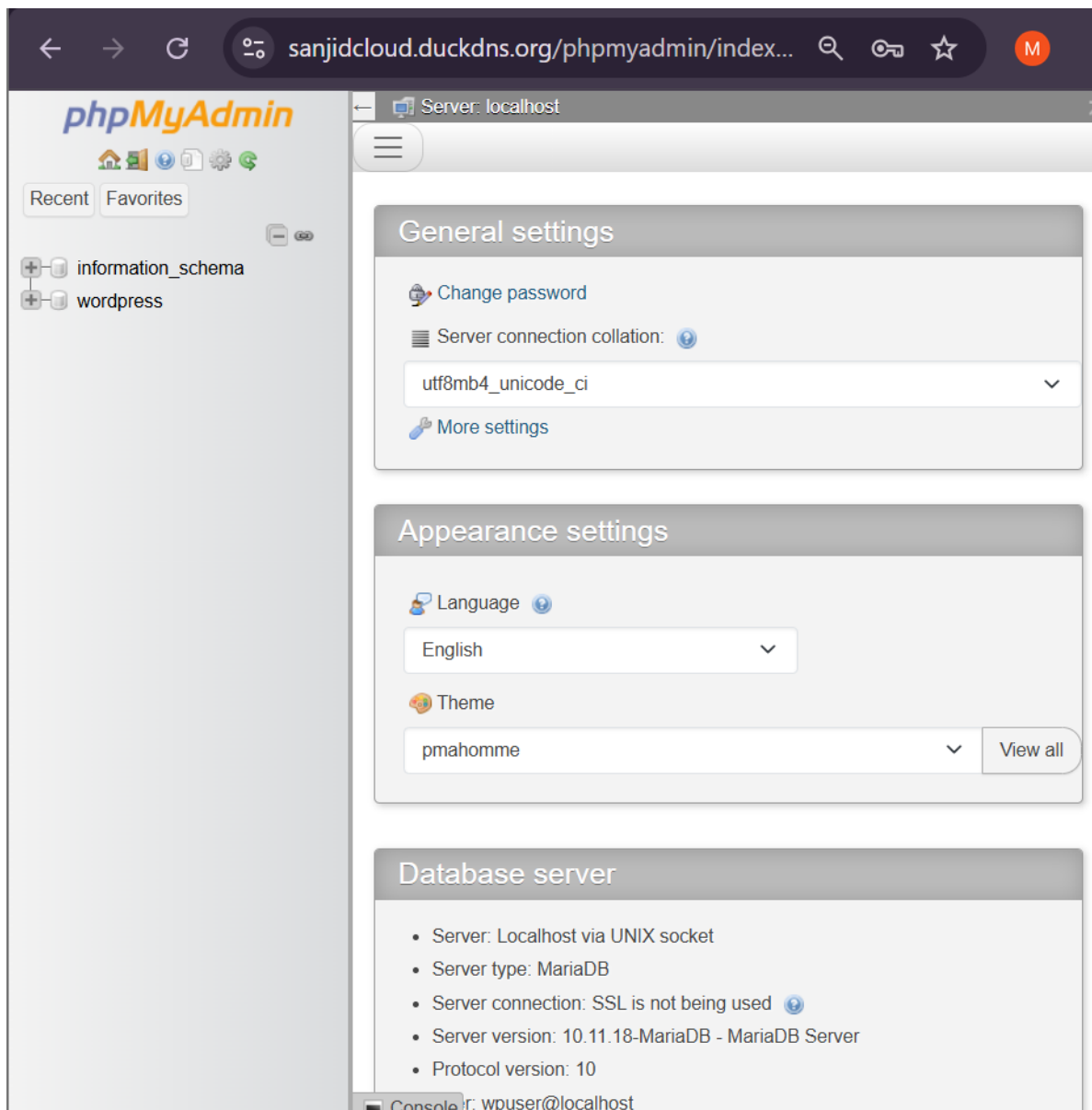
The directory contained index.php and the phpMyAdmin application files.

The phpMyAdmin URL was:

<https://sanjidcloud.duckdns.org/phpmyadmin/>



The screenshot shows a web browser window with the address bar displaying `sanjidcloud.duckdns.org/phpmyadmin/`. The page features the phpMyAdmin logo, which includes a stylized sailboat and the text "phpMyAdmin" and "Welcome to phpMyAdmin". Below the logo, there is a "Language" dropdown menu currently set to "English". Underneath, there is a "Log in" section with a blue circular icon containing a plus sign. This section contains two input fields: "Username:" and "Password:". At the bottom right of the login section is a "Log in" button. The browser's address bar also shows navigation icons (back, forward, refresh) and a search icon.



11. WordPress Installation

The wordpress role installed WordPress into the website root.

roles/wordpress/tasks/main.yml

- name: Install WordPress dependencies
 ansible.builtin.dnf:
 name:

```
- php-fpm
- php-mysqlnd
- php-gd
- php-mbstring
- php-xml
- php-curl
- php-zip
- tar
state: present

- name: Download WordPress
  ansible.builtin.get_url:
    url: https://wordpress.org/latest.tar.gz
    dest: /tmp/wordpress.tar.gz
    mode: "0644"

- name: Extract WordPress
  ansible.builtin.unarchive:
    src: /tmp/wordpress.tar.gz
    dest: /tmp
    remote_src: true

- name: Copy WordPress files to website root
  ansible.builtin.copy:
    src: /tmp/wordpress/
    dest: "{{ website_root }}"
    remote_src: true
    owner: "{{ website_user }}"
    group: "{{ website_user }}"
    mode: "0755"

- name: Create WordPress configuration
  ansible.builtin.template:
    src: wp-config.php.j2
    dest: "{{ website_root }}/wp-config.php"
    owner: "{{ website_user }}"
    group: "{{ website_user }}"
    mode: "0644"

- name: Set SELinux context for WordPress files
  ansible.builtin.command:
```

```
cmd: "restorecon -Rv {{ website_root }}"  
changed_when: false
```

WordPress was installed in:

/home/sanjid/mywebsite/public/

12. WordPress Database Configuration

The WordPress configuration template was:

roles/wordpress/templates/wp-config.php.j2

```
<?php  
define( 'DB_NAME', '{{ wordpress_db_name }}' );  
define( 'DB_USER', '{{ wordpress_db_user }}' );  
define( 'DB_PASSWORD', '{{ wordpress_db_password }}' );  
define( 'DB_HOST', '{{ wordpress_db_host }}' );  
define( 'DB_CHARSET', 'utf8mb4' );  
define( 'DB_COLLATE', '' );  
  
$table_prefix = 'wp_';  
  
define( 'WP_DEBUG', false );  
  
if ( ! defined( 'ABSPATH' ) ) {  
    define( 'ABSPATH', __DIR__ . '/' );  
}  
  
require_once ABSPATH . 'wp-settings.php';
```

The resulting configuration connected WordPress to:

Database: wordpress
User: wpuser
Host: localhost

The configuration file was created at:

/home/sanjid/mywebsite/public/wp-config.php

13. SSL Configuration

The `ssl` role was used to install the free Let's Encrypt certificate.

`roles/ssl/tasks/main.yml`

```
---
- name: Install Python pip and packaging
  ansible.builtin.dnf:
    name:
      - python3-pip
      - python3-packaging
    state: present

- name: Install Certbot with pip
  ansible.builtin.pip:
    name:
      - certbot
      - certbot-nginx
    executable: pip3

- name: Obtain Let's Encrypt certificate
  ansible.builtin.command:
    cmd: >
      certbot --nginx
      -d sanjidcloud.duckdns.org
      --non-interactive
      --agree-tos
      -m mhdsanjid07@gmail.com
      --redirect
    args:
      creates:
/etc/letsencrypt/live/sanjidcloud.duckdns.org/fullchain.pem

- name: Configure Nginx for HTTPS
  ansible.builtin.template:
    src: mywebsite.conf.j2
    dest: /etc/nginx/conf.d/mywebsite.conf
    owner: root
    group: root
```

```

mode: '0644'
notify: Restart Nginx
[ec2-user@ip-172-31-46-45 ~]$ sudo /usr/local/bin/certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

-----
Found the following certs:
Certificate Name: sanjidcloud.duckdns.org
Serial Number: 569355562b593d0bb230ac182c5249f8e9a
Key Type: ECDSA
Identifiers: sanjidcloud.duckdns.org
Expiry Date: 2027-01-04 05:51:17+00:00 (VALID: 89 days)
Certificate Path: /etc/letsencrypt/live/sanjidcloud.duckdns.org/fullchain.pem
Private Key Path: /etc/letsencrypt/live/sanjidcloud.duckdns.org/privkey.pem
-----
[ec2-user@ip-172-31-46-45 ~]$ |

```

14. Final Nginx HTTPS Configuration

The SSL role used:

roles/ssl/templates/mywebsite.conf.j2

```

server {
    listen 80;
    server_name sanjidcloud.duckdns.org;

    root /home/sanjid/mywebsite/public;
    index index.php index.html;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location /phpmyadmin/ {
        try_files $uri $uri/ /phpmyadmin/index.php?$query_string;
    }

    location ~ ^/phpmyadmin/(.+\.php)$ {
        include /etc/nginx/fastcgi_params;
        fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
        fastcgi_param PATH_INFO "";
        fastcgi_pass unix:/run/php-fpm/www.sock;
    }
}

```

```

        location ~ /\.php$ {
            include /etc/nginx/fastcgi_params;
            fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
            fastcgi_pass unix:/run/php-fpm/www.sock;
        }
    }

server {
    listen 443 ssl;
    server_name sanjidcloud.duckdns.org;

    root /home/sanjid/mywebsite/public;
    index index.php index.html;

    ssl_certificate
/etc/letsencrypt/live/sanjidcloud.duckdns.org/fullchain.pem;
    ssl_certificate_key
/etc/letsencrypt/live/sanjidcloud.duckdns.org/privkey.pem;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location /phpmyadmin/ {
        try_files $uri $uri/ /phpmyadmin/index.php?$query_string;
    }

    location ~ ^/phpmyadmin/(.+\.php)$ {
        include /etc/nginx/fastcgi_params;
        fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
        fastcgi_param PATH_INFO "";
        fastcgi_pass unix:/run/php-fpm/www.sock;
    }

    location ~ /\.php$ {
        include /etc/nginx/fastcgi_params;
        fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
        fastcgi_pass unix:/run/php-fpm/www.sock;
    }

```

```
}  
}
```

SSL Handler

```
---  
- name: Restart Nginx  
  ansible.builtin.service:  
    name: nginx  
    state: restarted
```

After applying the configuration, Nginx was listening on:

```
0.0.0.0:80  
0.0.0.0:443
```

HTTPS was tested using:

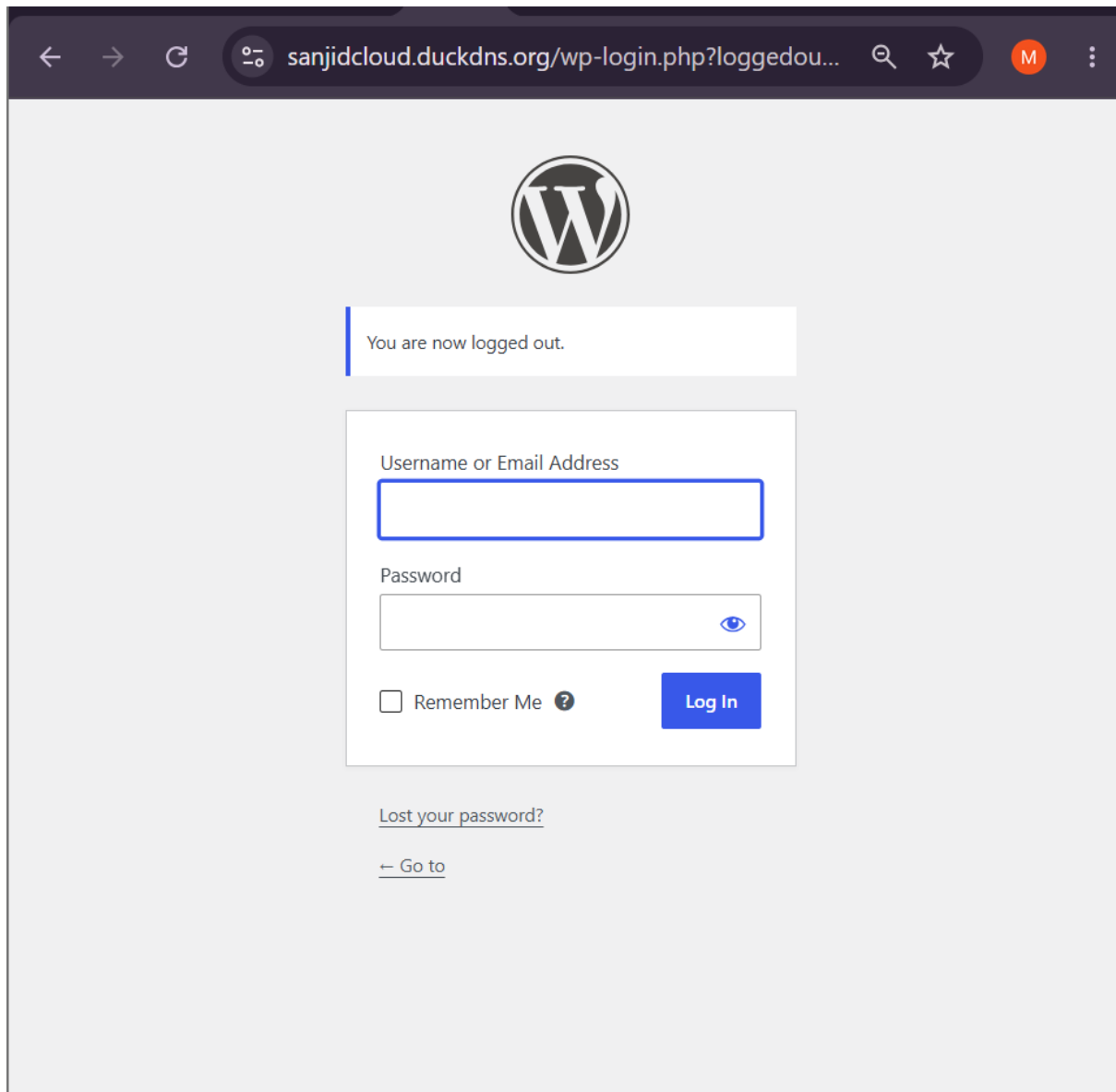
```
curl -Iv https://sanjidcloud.duckdns.org/
```

The certificate verification succeeded and the server returned:

```
HTTP/1.1 200 OK
```

The certificate was issued by Let's Encrypt for:

```
sanjidcloud.duckdns.org
```



15. Ansible Playbook Validation

Before applying the configuration, the playbook syntax was checked:

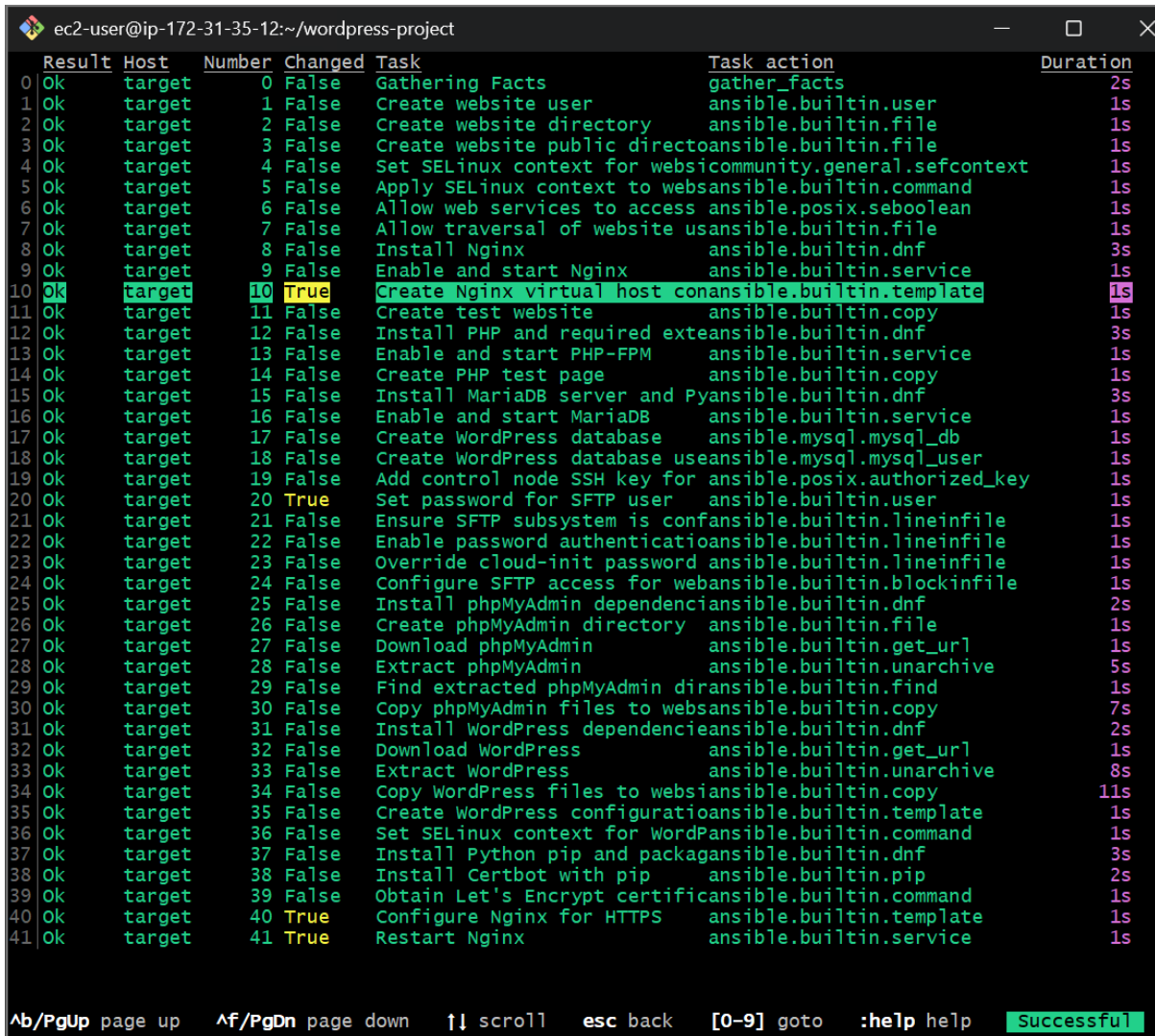
```
ansible-navigator run site.yml --ee false --syntax-check
```

The syntax check completed successfully.

The complete configuration was then applied:

```
ansible-navigator run site.yml --ee false
```

The playbook completed successfully and configured all required components.



	Result	Host	Number	Changed	Task	Task action	Duration
0	ok	target	0	False	Gathering Facts	gather_facts	2s
1	ok	target	1	False	Create website user	ansible.builtin.user	1s
2	ok	target	2	False	Create website directory	ansible.builtin.file	1s
3	ok	target	3	False	Create website public directory	ansible.builtin.file	1s
4	ok	target	4	False	Set SELinux context for website	community.general.sefcontext	1s
5	ok	target	5	False	Apply SELinux context to website	ansible.builtin.command	1s
6	ok	target	6	False	Allow web services to access website	ansible.posix.seboolean	1s
7	ok	target	7	False	Allow traversal of website directory	ansible.builtin.file	1s
8	ok	target	8	False	Install Nginx	ansible.builtin.dnf	3s
9	ok	target	9	False	Enable and start Nginx	ansible.builtin.service	1s
10	ok	target	10	True	Create Nginx virtual host configuration	ansible.builtin.template	1s
11	ok	target	11	False	Create test website	ansible.builtin.copy	1s
12	ok	target	12	False	Install PHP and required extensions	ansible.builtin.dnf	3s
13	ok	target	13	False	Enable and start PHP-FPM	ansible.builtin.service	1s
14	ok	target	14	False	Create PHP test page	ansible.builtin.copy	1s
15	ok	target	15	False	Install MariaDB server and Python connector	ansible.builtin.dnf	3s
16	ok	target	16	False	Enable and start MariaDB	ansible.builtin.service	1s
17	ok	target	17	False	Create WordPress database	ansible.mysql.mysql_db	1s
18	ok	target	18	False	Create WordPress database user	ansible.mysql.mysql_user	1s
19	ok	target	19	False	Add control node SSH key for WordPress	ansible.posix.authorized_key	1s
20	ok	target	20	True	Set password for SFTP user	ansible.builtin.user	1s
21	ok	target	21	False	Ensure SFTP subsystem is configured	ansible.builtin.lineinfile	1s
22	ok	target	22	False	Enable password authentication for SFTP	ansible.builtin.lineinfile	1s
23	ok	target	23	False	Override cloud-init password	ansible.builtin.lineinfile	1s
24	ok	target	24	False	Configure SFTP access for WordPress	ansible.builtin.blockinfile	1s
25	ok	target	25	False	Install phpMyAdmin dependencies	ansible.builtin.dnf	2s
26	ok	target	26	False	Create phpMyAdmin directory	ansible.builtin.file	1s
27	ok	target	27	False	Download phpMyAdmin	ansible.builtin.get_url	1s
28	ok	target	28	False	Extract phpMyAdmin	ansible.builtin.unarchive	5s
29	ok	target	29	False	Find extracted phpMyAdmin directory	ansible.builtin.find	1s
30	ok	target	30	False	Copy phpMyAdmin files to website	ansible.builtin.copy	7s
31	ok	target	31	False	Install WordPress dependencies	ansible.builtin.dnf	2s
32	ok	target	32	False	Download WordPress	ansible.builtin.get_url	1s
33	ok	target	33	False	Extract WordPress	ansible.builtin.unarchive	8s
34	ok	target	34	False	Copy WordPress files to website	ansible.builtin.copy	11s
35	ok	target	35	False	Create WordPress configuration file	ansible.builtin.template	1s
36	ok	target	36	False	Set SELinux context for WordPress	ansible.builtin.command	1s
37	ok	target	37	False	Install Python pip and packages	ansible.builtin.dnf	3s
38	ok	target	38	False	Install Certbot with pip	ansible.builtin.pip	2s
39	ok	target	39	False	Obtain Let's Encrypt certificate	ansible.builtin.command	1s
40	ok	target	40	True	Configure Nginx for HTTPS	ansible.builtin.template	1s
41	ok	target	41	True	Restart Nginx	ansible.builtin.service	1s

Ab/PgUp page up Af/PgDn page down ↑↓ scroll esc back [0-9] goto :help help Successful

16. WordPress Dashboard Configuration

After the server configuration was completed, the WordPress dashboard was accessed through:

<https://sanjidcloud.duckdns.org/wp-admin/>

The WordPress administrator account was:

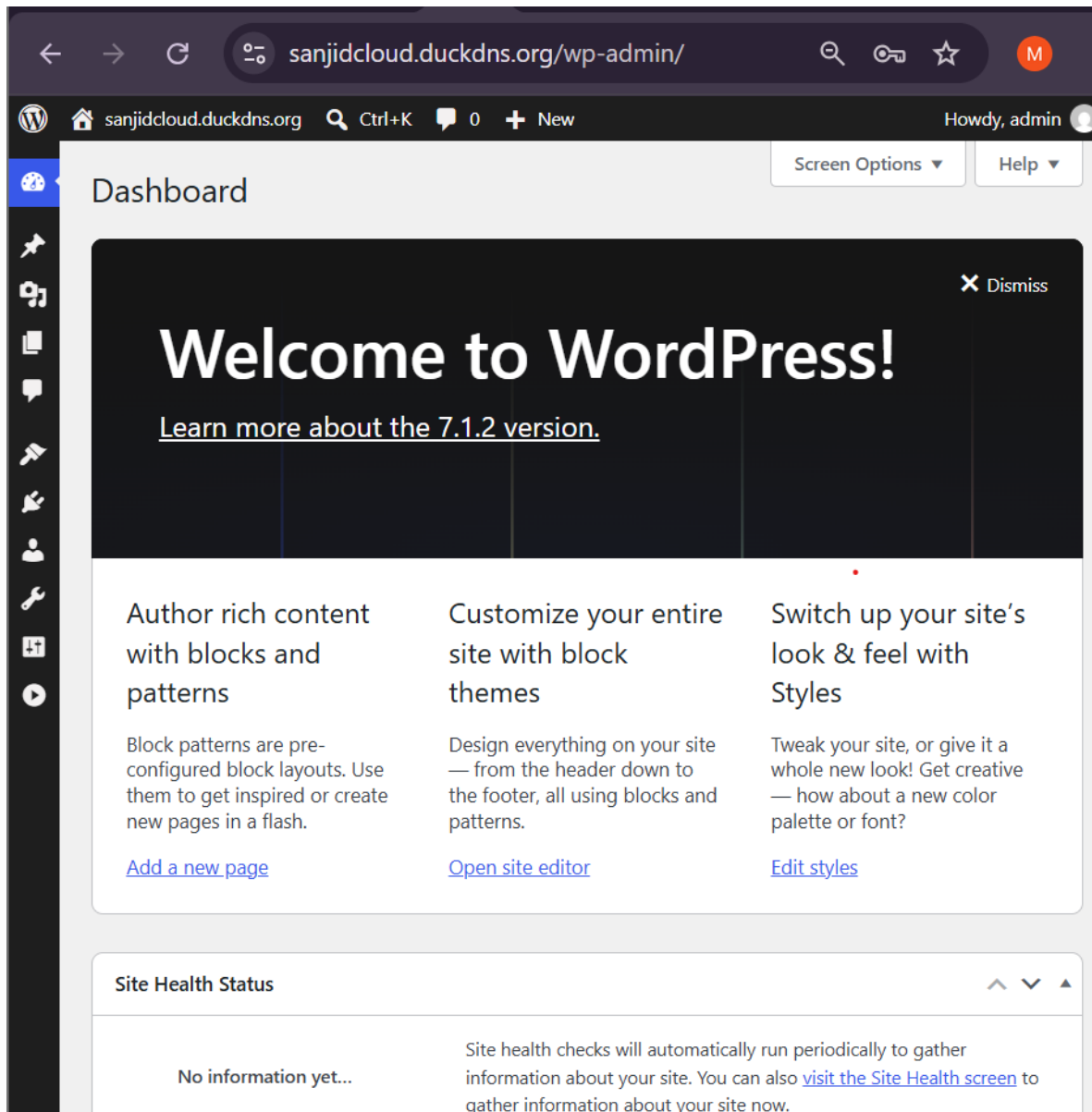
Username: admin

Password: 123

The WordPress dashboard was successfully accessed.

The installed WordPress version shown in the dashboard was:

7.1.2



17. Personal Blog Post

A personal blog post was created through the WordPress dashboard.

The title of the post was:

About me

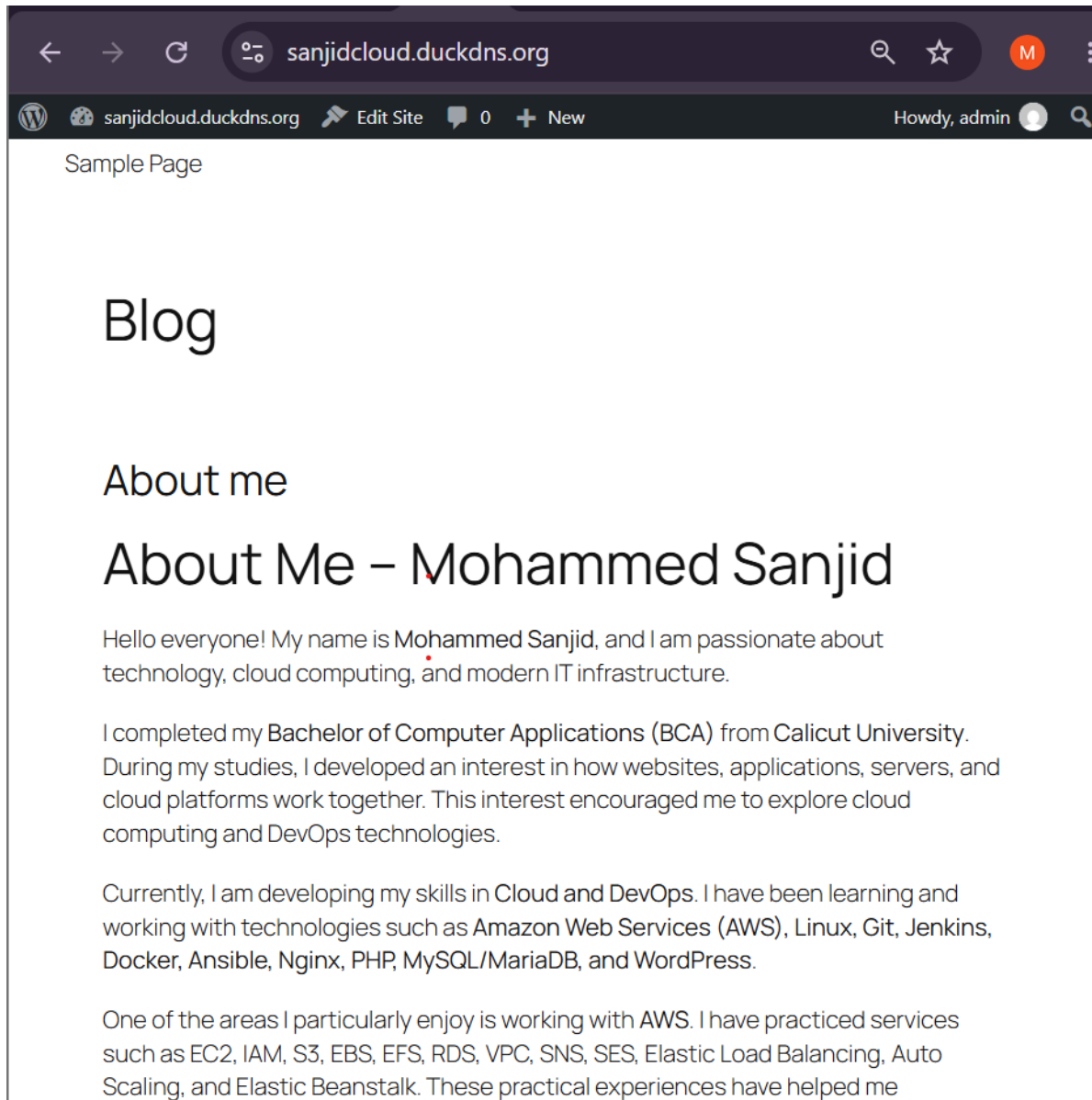
The post was created by:

admin

The post was published successfully on:

October 6, 2026

The published post was visible from the WordPress Posts section.



18. Verification

The following components were verified after the Ansible deployment:

Website

<https://sanjidcloud.duckdns.org/>

The website loaded successfully.

PHP

PHP 8.3.33 was installed and PHP-FPM was running.

MariaDB

MariaDB 10.11.18 was running with:

Database: wordpress

User: wpuser

Website Directory

/home/sanjid/mywebsite/public

was used as the website root.

SFTP

Password-based SFTP login was successfully tested.

The sanjid user was restricted to SFTP and could not obtain a normal SSH shell.

phpMyAdmin

phpMyAdmin was accessible at:

<https://sanjidcloud.duckdns.org/phpmyadmin/>

SSL

HTTPS was successfully configured using a Let's Encrypt certificate.

Nginx was listening on:

80

443

WordPress

WordPress was successfully installed and the administrator dashboard was accessible.

Blog

The personal post “**About me**” was successfully published.

19. Credentials

SFTP Credentials

Host: sanjidcloud.duckdns.org

Username: sanjid

Password: [SFTP PASSWORD]

phpMyAdmin Credentials

URL: <https://sanjidcloud.duckdns.org/phpmyadmin/>

Username: wpuser

Password: [DATABASE PASSWORD]

WordPress Credentials

WordPress Admin URL: <https://sanjidcloud.duckdns.org/wp-admin/>

Username: admin

Password: [WORDPRESS ADMIN PASSWORD]

20. Complete Configuration Summary

Component	Configuration
Control Node	RHEL 10.2
Target Server	RHEL 10.2
Automation	Ansible
Ansible Navigator	26.9.0
Web Server	Nginx 1.26.3
PHP	8.3.33
Database	MariaDB 10.11.18
Website User	sanjid
Website	mywebsite
Website Root	/home/sanjid/mywebsite/public
Domain	sanjidcloud.duckdns.org
SFTP	OpenSSH Internal SFTP
phpMyAdmin	/phpmyadmin/
WordPress	Installed
WordPress Database	wordpress
WordPress DB User	wpuser
SSL	Let's Encrypt
HTTP	Port 80
HTTPS	Port 443
WordPress Admin	admin
Personal Post	About me

Conclusion

The complete WordPress hosting environment was configured using **Ansible roles and the main site.yml playbook**. The implementation included the website user and directory structure, Nginx, PHP/PHP-FPM, MariaDB, SFTP, phpMyAdmin, WordPress and Let's Encrypt SSL.

All individual roles were included in the final playbook, and the playbook was syntax-checked and executed successfully. SFTP access was tested using username and password authentication and was restricted to SFTP-only access. phpMyAdmin was made accessible through the website, and the WordPress application was connected to MariaDB.

The domain `sanjidcloud.duckdns.org` was configured with a valid Let's Encrypt certificate, and HTTPS was verified successfully. WordPress was then accessed

through the administrator dashboard, and the personal blog post “**About me**” was created and published.